

Reviewer Pack — Symmetric Group Character Eigenvalue Lower Bound on SOS Refu...

Entry 1620c2565d7f · INCONCLUSIVE

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Symmetric Group Character Eigenvalue Lower Bound on SOS Refutation Degree

Entry ID: 1620c2565d7f **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-20 06:00:45 UTC

1. Conjecture

Field A (mathematical branch): Representation Theory of Symmetric Groups **Field B** (complexity object): SOS Refutation Degree for 3-CNF Formulas

Statement:

For a 3-CNF formula φ on n variables, let χ be the character matrix of the symmetric group S_n induced by its clause permutation action. The SOS refutation degree of φ is at least the largest eigenvalue of χ , denoted $\lambda_{\max}(\chi)$.

Rationale (proposer's reasoning):

Symmetric group representations capture combinatorial symmetries of CSPs. The eigenvalues of character matrices encode structural constraints that may limit the efficiency of SOS hierarchies, as eigenvalues reflect the 'spread' of group actions on constraint satisfaction.

Taxonomy category: SOS_HIERARCHY (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 63cd00e2ffb57043

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.95	UNCERTAIN	SAFE
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def generate_3cnf(n, m):
    clauses = set()
    while len(clauses) < m:
        clause = tuple(random.sample(range(1, n+1), 3))
        if clause not in clauses and -clause not in clauses:
            clauses.add(clause)
    return clauses

def random_permutation(n):
    perm = list(range(n))
    random.shuffle(perm)
    return perm

def character_matrix(n, m, clauses):
    chi = [[0] * n for _ in range(n)]
    for clause in clauses:
        perm = random_permutation(n)
        for i in range(n):
            chi[perm[i]][i] += 1
    return chi

def largest_eigenvalue(chi):
    n = len(chi)
    eigenvalues = [0] * n
    for _ in range(100): # Power iteration method
        v = [random.random() for _ in range(n)]
        v /= math.sqrt(sum(x**2 for x in v))
        v_next = [sum(chi[i][j] * v[j] for j in range(n)) for i in range(n)]
        v_next /= math.sqrt(sum(x**2 for x in v_next))
        eigenvalues[_ % n] = sum(v_next[i] * v[i] for i in range(n))
    return max(eigenvalues)

def sos_refutation_degree(cnf):
    # Placeholder function to simulate SOS refutation degree computation
    return len(cnf) # Simplified for testing purposes
```

```

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = random.randint(5, 40)
    m = random.randint(n * (n - 1), n * (n + 1))
    cnf = generate_3cnf(n, m)
    chi = character_matrix(n, m, cnf)
    lambda_max = largest_eigenvalue(chi)
    degree = sos_refutation_degree(cnf)

    return {
        "metric_name": "SOS Refutation Degree Lower Bound",
        "metric_value": lambda_max,
        "instances_tested": 1,
        "conjecture_holds": lambda_max <= degree,
        "counterexample": "" if lambda_max <= degree else f"n={n}, m={m}"
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i - 1 for i in range(5, 30)]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_value = math.sqrt(sum((result["metric_value"] - mean_value)**2 for result in results) / len(results))
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not result["conjecture_holds"] for result in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\", first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE insufficient data")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_adf7e0c6.py", line 77, in <module>

```

    trial_result = run_trial(seed)
    ~~~~~
```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_adf7e0c6.py", line 58, in run_trial

```

    cnf = generate_3cnf(n, m)
    ~~~~~
```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_adf7e0c6.py", line 22, in generate_3cnf

if clause not in clauses and -clause not in clauses:

^^^^^^
TypeError: bad operand type for unary -: 'tuple'

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test crashed due to TypeError, preventing data collection. Support condition cannot be evaluated without valid trial results. | next: Fix the tuple handling in generate_3cnf() to resolve the TypeError and rerun tests

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_local	qwen3:8b	0	0	214703
2	preregistration	ollama_remote	qwen3:8b	0	0	31683
3	novelty	ollama_remote	qwen3:8b	0	0	27389
4	novelty	ollama_remote	qwen3:8b	0	0	19494
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13622
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16618
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10934
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9227
9	judge	ollama_remote	qwen3:8b	0	0	19222

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 362892 ms total latency. Provider mix: {'ollama_local': 1, 'ollama_remote': 8}

(full prompt+response transcripts available in `research/audit/1620c2565d7f.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/1620c2565d7f.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/1620c2565d7f.tar.gz` (if generated)